

Project: Smart Parking Platform

Version: 3.1

Date: 09.24.2026

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document defines the initial requirements for the Smart Parking Platform developed by WE ARE Software Corp. The system will help drivers locate, reserve, and pay for parking spaces in real time while providing parking operators with tools to manage their facilities and monitor occupancy. The platform will be available through web and mobile applications and will provide parking availability information, navigation assistance, and management tools for parking operators. The goal is to develop a project management plan for the creation of this system, including requirements gathering, stakeholder management, scheduling, budgeting, risk management, and team coordination.

1.2 Scope

The project will provide web and mobile applications that support parking searches, real-time availability, reservations, digital payments, navigation, notifications, and parking management. The project will also provide an administrative dashboard for parking operators to manage facilities and view occupancy information and reports. Physical parking garage construction, manufacturing parking hardware, and direct control of city traffic systems are out of scope.

2. Overall Description

2.1 Users:

- Vehicle Owners
- Parking Facility Owners/Employees
- Finance and Billing Employees
- Third-Party Payment Providers
- City Transportation Employees
- Mobile Phone Users

2.2 System Environment:

The Smart Parking Platform will consist of web-based and mobile applications connected to backend/API services and a database. The system will also integrate with external mapping/navigation and payment services.

2.3 Constraints:

- The project has a fixed semester timeline.
- The project has an unlimited theoretical development budget.
- The system depends on third-party mapping, navigation, payment, and other APIs.
- The system must follow applicable security and privacy requirements.
- The system must consider applicable city ordinances and laws.
- Users must authenticate to access protected account and management functions.

2.4 Assumptions:

- Users have internet access.
- Users have a compatible web browser or mobile device.
- Parking facilities provide accurate availability and occupancy information.
- Third-party mapping and payment services are available when required.
- Parking operators will provide and maintain accurate facility information.

3. Functional Requirements

FR1: The system shall allow users to register and log in securely.

FR2: The system shall allow users to search for available parking locations.

FR3: The system shall display parking locations and availability on an interactive map.

FR4: The system shall allow users to reserve available parking spaces.

FR5: The system shall process digital parking payments through a supported payment provider.

FR6: The system shall provide users with reservation history and receipts.

FR7: The system shall send users reservation-related notifications and alerts.

FR8: The system shall provide navigation assistance to selected parking facilities through an external mapping service.

FR9: The system shall allow parking operators to add and manage parking facilities.

FR10: The system shall allow parking operators to monitor parking occupancy.

FR11: The system shall provide occupancy and parking usage reports.

FR12: The system shall allow authorized users to manage relevant account and parking information.

Additional functional requirements will be identified through stakeholder analysis and requirements elicitation throughout the semester.

4. Non-Functional Requirements

NFR1: The system shall support at least 1,000 concurrent users.

NFR2: The system shall display parking availability updates within 1 second when current data is available.

NFR3: The system shall target 99.99% uptime for critical services.

NFR4: The system shall protect user account, vehicle, reservation, and payment-related information.

NFR5: The system shall support modern web browsers and supported mobile devices.

NFR6: The system shall be scalable to support additional users and parking facilities.

NFR7: The system shall maintain consistent reservation and payment information when transactions are completed successfully.

5. Use Cases

Use Case 1

Title: Reserve a Parking Space

Actor: Driver

Precondition: User is logged in and a parking facility has an available space.

Steps:

- User searches for a parking location.

- User selects an available parking facility.
- User selects the reservation date and time.
- System verifies parking availability.
- User confirms the reservation and completes payment.
- System displays the reservation confirmation.

Postcondition: The parking reservation is saved to the user's account.

Work Breakdown Structure (WBS)

1. Smart Parking Platform

1.1 User & Account Management

1.1.1 User Registration & Login

Account creation

Secure login

1.1.2 User Profile

Account information

Vehicle information

1.1.3 Authentication & Security

Authentication controls

Protected user data

1.2 Parking Search & Availability

1.2.1 Parking Location Search

Search available facilities

Filter parking locations

1.2.2 Real-Time Availability

Display available spaces

Update occupancy information

1.2.3 Interactive Parking Map

Display parking locations

Select parking facility

1.3 Reservations

1.3.1 Space Selection

Select parking facility

Select available space

1.3.2 Reservation Management

Select date and time

Confirm reservation

1.3.3 Reservation History

View past reservations

View reservation details

1.4 Payments

1.4.1 Payment Processing

- Connect payment provider

- Process parking payment

1.4.2 Payment Confirmation

- Confirm successful payment

- Store transaction information

1.4.3 Receipts

- Generate digital receipt

- View receipt history

1.5 Navigation & Notifications

1.5.1 Navigation Integration

- Connect mapping service

- Provide directions to facility

1.5.2 User Notifications

- Reservation confirmation

- Reservation alerts

1.6 Parking Operator Management

1.6.1 Facility Management

- Add parking facility

- Update facility information

- Remove facility

1.6.2 Occupancy Management

- Monitor occupancy

- Update availability

1.6.3 Operator Dashboard

- View facility information

- View parking activity

1.7 Reporting & Analytics

1.7.1 Occupancy Reports

- Generate occupancy reports

- View occupancy trends

1.7.2 Parking Usage Reports

- Analyze parking usage

- Review reservation activity

1.8 Testing & Deployment

1.8.1 System Testing

- Functional testing

- Integration testing

1.8.2 Security & Performance Testing

- Security testing

- Performance testing
- 1.8.3 Deployment
 - Deploy web application
 - Deploy mobile application
 - Deploy backend services

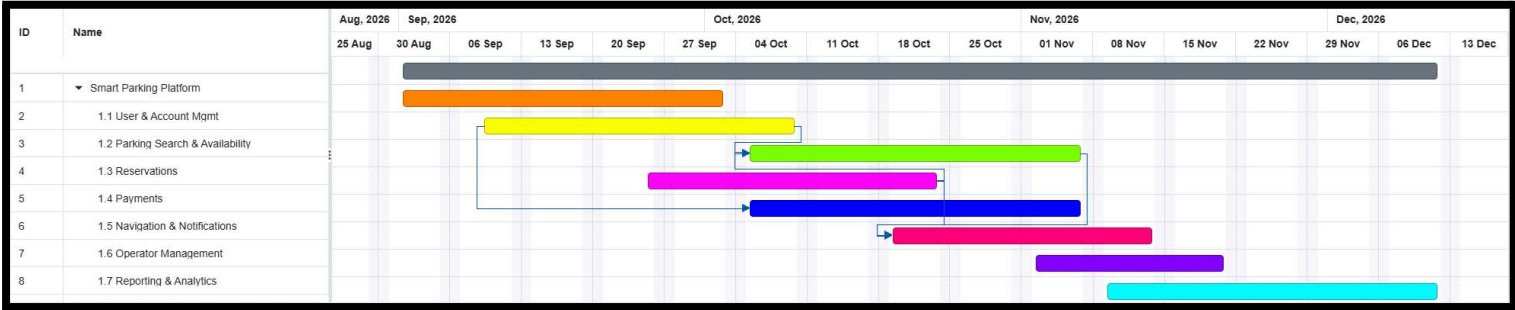
Project Timeline

WBS Component	Story Points	Risk
1.1 User & Account Mgmt	8	Moderate complexity; relies on secure auth standards
1.2 Parking Search & Availability	5	Depends on third-party map API reliability
1.3 Reservations	13	High complexity; requires state management & concurrency handling
1.4 Payments	8	Standard gateway integration but needs error/retry logic
1.5 Navigation & Notifications	3	Lightweight API calls; mostly configuration-driven
1.6 Operator Management	5	CRUD-heavy but straightforward UI/backend mapping
1.7 Reporting & Analytics	8	Requires data aggregation & charting libraries
1.8 Testing & Deployment	5	Cross-team coordination; environment setup overhead

WBS Component	Start	End	Duration	Dependencies
1.1 User & Account Mgmt	Sept Week 1	Oct Week 1	~5 wks	–
1.2 Parking Search & Availability	Sept Week 2	Oct Week 2	~5 wks	–
1.3 Reservations	Oct Week 1	Nov Week 1	~5 wks	1.2 (Finished)
1.4 Payments	Sept Week 4	Oct Week 3	~5 wks	1.4 (Started)
1.5 Navigation & Notifications	Oct Week 2	Nov Week 1	~5 wks	1.3 (Started)
1.6 Operator Management	Oct Week 3	Nov Week 2	~4 wks	1.3 (Finished~), 1.4 (Started)

1.7 Reporting & Analytics	Nov Week 1	Nov Week 3	~3 wks	Backend API Calling
1.8 Testing & Deployment	Nov Week 2	Dec Week 2	~5 wks	First working Dev component required to start testing

Gantt Chart



Milestones

Milestone	Target Date	Success Criteria
Requirements & WBS Baseline	Sept 7 (Week 1)	SRS v1.0 approved, WBS signed off by team leads
Sprint 1: Core Search & Auth Complete	Oct 9 (Week 6)	Users can log in, search lots, view live availability
Sprint 2: Reservation & Payment Flow Live	Nov 6 (Week 10)	End-to-end booking + payment confirmation working
Sprint 3: Operator Dashboard & Reports Ready	Nov 27 (Week 13)	Facilities manageable, occupancy/reports viewable
Sprint 4: Final Testing and Product Release	Dec 11th (Week 15)	Product Testing Completed, Product Released to public and team reorganized to focus on maintenance

Product Backlog

Login

1	User Registration	Sprint 1
2	User Login	Sprint 1
3	Secure Password Handling	Sprint 1
4	User Logout	Sprint 1
5	Role-based Access for Users and Operators	Sprint 1
6	Authentication Testing	Sprint 1
7	Password Reset	Sprint 2

User/Operator UI

1	User Registration Screen	Sprint 1
2	User Login Screen	Sprint 1
3	User Dashboard	Sprint 1
4	User Profile Screen	Sprint 1
5	Operator Dashboard	Sprint 1
6	Vehicle Information Screen	Sprint 2
7	Parking Search Screen	Sprint 2
8	Interactive Parking Map	Sprint 2
9	Parking Availability Display	Sprint 2
10	Parking Facility Details	Sprint 2
11	Operator Facility Management Screen	Sprint 2
12	Parking Reservation Screen	Sprint 3
13	Payment Screen	Sprint 3

14	Reservation Confirmation Screen	Sprint 3
15	Reservation Confirmation Notification	Sprint 3
16	Upcoming Reservation Reminder	Sprint 4
17	Reservation Cancellation Notification	Sprint 4
18	Reservation History Screen	Sprint 4
19	Payment History Screen	Sprint 4
20	Digital Receipt History Screen	Sprint 4

User/Operator Backend

1	User Account API	Sprint 1
2	Authentication API	Sprint 1
3	User Profile API	Sprint 1
4	Vehicle Information API	Sprint 2
5	Parking Facility API	Sprint 2
6	Parking Search API	Sprint 2
7	Real-time Availability API	Sprint 2
8	Occupancy Update API	Sprint 2
9	Operator Facility Management API	Sprint 2
10	Update Parking Facility Information	Sprint 2
11	Remove Parking Facility	Sprint 2
12	Update Parking Space Availability	Sprint 2
13	Integrate External Mapping Service	Sprint 2
14	Secure API Access and Authorization	Sprint 2
15	Reservation API	Sprint 3
16	Reservation Availability Validation	Sprint 3
17	Reservation Cancellation API	Sprint 3

18	Payment Processing Integration	Sprint 3
19	Payment Confirmation Processing	Sprint 3
20	Prevent Double Booking of Parking Spaces	Sprint 3
21	Handle Failed Payment Transactions	Sprint 3
22	Operator Occupancy Monitoring	Sprint 3
23	Provide Navigation to Selected Parking Facility	Sprint 3
24	Receipt Generation Service	Sprint 4
25	Reservation Cancellation and Refund Processing	Sprint 4
26	Record account and System Activity Logs	Sprint 4
27	Payment Integration Testing	Sprint 4
28	Final System Integration Testing	Sprint 5

Reporting Frontend/Backend

1	Reservation History	Sprint 3
2	Payment History	Sprint 3
3	Occupancy Report	Sprint 4
4	Daily Occupancy Report	Sprint 4
5	Weekly Occupancy Report	Sprint 4
6	Parking Usage Report	Sprint 4
7	Reservation Activity Report	Sprint 4
8	Facility Utilization Report	Sprint 5
9	Revenue/Payment Report	Sprint 5
10	Performance Testing for Concurrent Users	Sprint 5